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Secrecy Performance Analysis of Energy Harvesting Wireless Sensor Networks With a Friendly Jammer

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ABSTRACT The broadcast nature of energy harvesting wireless sensor networks (EH-WSNs) allows sensor nodes (SNs) within the coverage range of a transmitter to capture its signals. However, an EH-WSN is vulnerable to eavesdropping and signal interception; therefore, security in the EH-WSNs is of significant interest, and this issue has been addressed over many years. However, no work has studied the existence of a friendly jammer to mitigate the security impact. Thus, this paper proposes a model and optimization scheme that uses a wirelessly powered friendly jammer to improve secrecy in EH-WSNs. The considered EH-WSN model includes multiple power stations, multiple SNs (sources) and their base station, a friendly jammer, and multiple passive eavesdroppers. We divide the model into two phases: 1) the power stations transfer RF energy to the source SNs and 2) the source SNs transmit information to their base station, while a friendly jammer generates jamming signals against multiple eavesdroppers. Using statistical characteristics of the signal-to-noise ratio, the closed-form expressions of the existence probability of the secrecy capacity and secrecy outage probability are derived. We also propose an optimal sensor scheduling scheme to enhance physical layer secrecy (i.e., best-node scheduling), and we demonstrate our method's superior performance compared with a conventional round-robin scheduling scheme. The analysis of the simulation results supports our hypothesis, which is in line with Monte Carlo simulations.

INDEX TERMS Energy harvesting, wireless sensor networks, friendly jammer, physical layer secrecy.

I. INTRODUCTION

The Internet of Things (IoT) has attracted several researchers' attention worldwide. IoT is an intelligent infrastructure comprising (a huge number of) uniquely identifiable devices, services, and people capable of wirelessly communicating with each other through the Internet [1], [2]. The main driving idea for the future of IoT is related to smart sensors and other communication technologies, including wireless sensor networks (WSNs), nanotechnology, and miniaturization. WSNs present extensive potential in future applications, including security and tactical surveillance, intrusion detection, disaster management, weather monitoring, inventory control, traceability, animal tracking, and health care services [3].

However, one of the greatest challenges faced by WSNs is energy constraints: sensor devices have little memory, small batteries, and tiny central processing units [4]. To extend the

networking coverage, particularly for wireless communications, certain communication protocols are required, and their complexities may increase the energy consumption of the sensor nodes (SNs), resulting in reduced overall network lifetime. Hence, various mechanisms such as energy-aware MAC protocols, power-aware storage, routing and data dissemination protocols, duty-cycling strategies, adaptive sensing rates, tiered system architectures, and redundant placements of SNs have been proposed for SNs to cause the networks to consume less energy. Although all the above techniques optimize and adapt energy usage to maximize the lifetime of the SNs, their lifetime remains limited, which may restrict SN operations over the long term [5].

A recent technology trend in energy harvesting provides a fundamental method to address the problem of finite SN lifetime and constitutes a promising approach for the

emerging IoT [6]. Energy harvesting in WSNs (EH-WSNs) refers to harvesting energy from the environment (e.g., via thermal, solar, vibration, and wireless Radio Frequency (RF) energy sources) and converting it into electrical energy for the SNs [7]. Although energy harvesting from the above environmental sources is dependent on the presence of the energy sources and must consider their unstable natures, RF energy harvesting may nevertheless represent a practical trend for future energy-aware systems because it is readily available in the form of transmitted energy (TV/radio broadcasts, mobile base stations, and handheld radios), has low costs, and is available in small form factor implementations. Note that the power conversion efficiency (PCE) of an RF converter can be quite high, up to 70 percent or more [8]–[10].

However, in addition to having some of the aforementioned advantages, RF EH-WSNs are often vulnerable to eavesdropping and signal interception due to the broadcast nature of wireless communications [11]. For example, situations may occur in which the SNs act as eavesdroppers to overhear messages transmitted from the source SNs to their sink [12], which presents a challenging security issue. Some research has focused on using cryptographic methods to ensure the security of the sensor-sink transmissions against eavesdropping [13]. However, these methods face some crucial challenges in performing key distribution/exchange between the SNs and the sink [14]. In addition, the use of data encryption/decryption algorithms introduces communication latency, high computational loads, and signaling overhead [15], all of which are unfavorable to WSNs.

To mitigate this problem, one potential approach, physical layer secrecy in wireless communications, has become a fascinating subject due to its low complexity, low latency, and ability to be combined with other security mechanisms to enhance overall information security [16], [17]. Physical layer secrecy was exploited by Shannon [18] and extended by Wyner [19], who established an information theoretic framework with a classical model consisting of one source and one destination in the presence of an eavesdropper. This method defined secrecy capacity as the difference between the channel capacity of the main link from the source to the destination and a wiretap link from the source to an eavesdropper [20].

Note that when the secrecy capacity is positive (i.e., the channel capacity of the main link is higher than that of the wiretap link) the system is protected, and the eavesdropper cannot capture the source-destination communication packets [21], [22]. This implies that the legitimate channel capacity can effectively decrease the probability that the eavesdropper will successfully overhears the packet. However, the secrecy capacity of wireless sensor transmissions is severely limited by wireless fading effects and metallic obstacles, which produce substantial changes and result in a further reduction in secrecy capacity.

To overcome this limitation, in this paper, we propose a new model and an optimization scheme to enhance the

secrecy performance for EH-WSNs. Our contributions are as follows:

- We propose a new WSN model with RF energy-harvesting SNs in which jamming is deployed in a cooperative manner to enable a secure communication link between a legitimate transmitter and an intended receiver to enhance the physical layer secrecy with an awareness of energy harvesting.
- We investigate an optimization scheme to protect EH-WSNs against eavesdropping attacks, that selects the SN with the highest gain to transmit its sensed information to the base station.
- We evaluate the system secrecy performance by deriving a closed-form expression for the existence probability of the secrecy capacity (PCS) and secrecy outage probability (SOP) that uses the statistical characteristics of the signal-to-noise ratio (SNR). This approach enables us to determine the main factors affecting the secrecy performance in high-SNR regimes.

The remainder of this paper is organized as follows: Section II presents a brief survey of the related work on physical layer security in EH-WSNs. Section III provides details concerning the system model, communication technique, and information transmission. In Section IV, the channel capacity, PCS, and SOP for the round-robin scheduling and the best-node scheduling schemes are analyzed. Section V provides numerical results and discussions. Finally, concluding remarks are given in Section VI.

II. RELATED WORK

With the rapid growth of IoT, WSNs have been studied extensively in recent years, with particular focus on aspects such as energy harvesting and the physical layer secrecy problem. Thus, in this section, we briefly summarize the related works concerning these two aspects for WSNs.

Radio frequency energy harvesting wireless networks are separated into two main families: simultaneous wireless information and power transfer (SWIPT) and wireless powered communication networks (WPCN) [23], [24]. In SWIPT, the transmitted signals carry both energy and packets to contemporaneously achieve information delivery and wireless energy recharging [25]. In contrast, WPCNs divide wireless communication into two phases. The power transfer phase first broadcasts energy-containing signals to recharge the energy harvesting SNs; then, these SNs transmit packets by utilizing the energy harvested in the previous phase.

WPCN technology operates under a two-phase harvest and transmission protocol; hence, it increases the network lifetime of the wireless devices [26]. Therefore, many novel studies have investigated physical layer secrecy in WPCN to maximize system performance. In 2016, Lee *et al.* [27] investigated the problem of maximizing the sum throughput for multi-user WPCNs. Simultaneously, Liu *et al.* [28] investigated a system model with a sink and an EH SN that harvests energy from environment sources such as solar,

wind, vibrations, and RF signals. This EH SN can monitor the status of certain properties, and it adopts WPCN to perform sensing and transmission. Then, they considered the energy costs of both sensing and transmission using two metrics: update age and update cycle. However, these works proposed only an optimal energy and time allocation algorithm without considering the network's security performance.

Zou and Wang [29] considered a model that included a sink and multiple SNs subject to a passive eavesdropping attacker. They evaluated the secrecy performance in two cases, round-robin scheduling and optimal sensor scheduling, using a closed-form expression of the probability of the occurrence of an intercept event (called the intercept probability). However, this work investigated only the ideal case, in which perfect channel state information (CSI) is available for the eavesdropping channel. Hence, Jameel *et al.* [15] also considered this model but focused on SOP under outdated CSI of the eavesdropper. Note that both studies investigated a simplified model; therefore, Deng *et al.* [30] proposed a three-tier model consisting of the sensing field, access points, and sinks. They investigated the impacts of random locations and spatial densities of the SNs, access points, sinks, and external eavesdroppers on the secrecy performance. Based on that investigation, they provided guidelines for secure transmission in practical WSNs as well as a deployment design. However, that work considered only the average secrecy rate metric without investigating secrecy outages in the system.

Accordingly, Zhu *et al.* [12] presented the challenges in and solutions for improving the physical layer security for Industrial Wireless Sensor Networks (IWSNs). They discussed some wireless attack techniques, including jamming and eavesdropping, along with their defense methods, where the jammers attempt to send an interfering signal to disrupt the communications between the SNs and their sink, while the eavesdroppers attempt to capture the radio signals of the communication sensor-sink transmissions. Based on that discussion, they investigated the model with SNs and their sink while considering an eavesdropper and evaluated the relationship between the outage probability and the intercept probability based on the outage probability expression of the sensor-sink transmission.

To the best of our knowledge, physical layer secrecy for EH-WSNs has not been clarified when considering a friendly jammer. Thus, one novel aspect of this research is the investigation of the secrecy performance of EH-WSNs with a friendly jammer and multiple eavesdroppers who attempt to wiretap the communications between the EH SNs and their sink.

III. SYSTEM AND CHANNEL MODEL

A. SYSTEM MODEL

Let us consider EH-WSNs for which power stations (PSs) P_n , for $n = 1, 2, \dots, N$, may be available in the form of transmission towers in an environment, such as TV broadcast towers (airwaves) [31], wireless local area networks (WLANs), or a global system for mobile

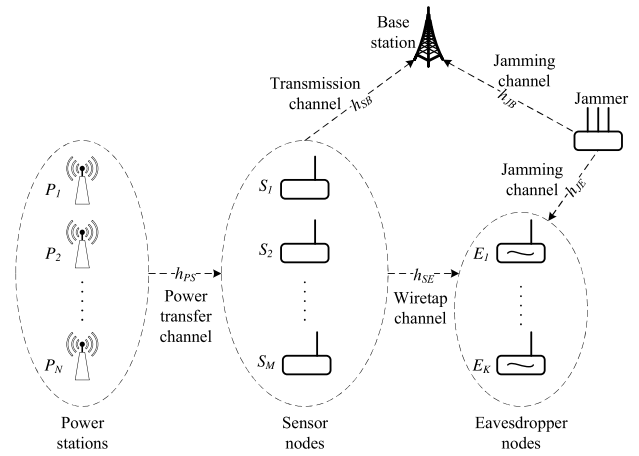


FIGURE 1. A system model of EH-WSNs with a friendly jammer. The SNs harvest the energy from multiple PSs and utilize this harvested energy to transfer packets to the BS. The SNs are overheard by multiple ENs.

communication (GSM) [8]. These operate on orthogonal frequency bands (i.e., they do not interfere with each other). The source SNs S_m , for $m = 1, 2, \dots, M$, harvest energy from P_n and then use this harvested energy to send information to the base station (BS) B in the presence of multiple passive eavesdroppers (ENs) E_k , for $k = 1, 2, \dots, K$, who wish to overhear the information transmitted from the SNs to the BS. The friendly jammer J interferes with these ENs, as illustrated in Fig. 1. The channel coefficients of the P_n -to- S_m , S_m -to- B , and J -to- B communication links are denoted by $h_{P_n S_m}$, $h_{S_m B}$, and h_{JB} , respectively. The channel gains of the S_m -to- E_k illegitimate links and the J -to- E_k interference links are expressed by $h_{S_m E_k}$ and h_{JE_k} , respectively. We assume that all channels undergo Rayleigh fading and that all nodes are equipped with a single antenna.

B. COMMUNICATION TECHNIQUE

The basic idea of the considered RF EH-WSNs is that the SNs can convert the PSs' emitted power into useful energy for the SNs' communications. Here, we considered multiple-source WPCNs [32] with a jammer and multiple ENs that attempt to wiretap the communication between the SNs and the BS. We adopt the harvest-use (HU) architecture [6]. Note that once an SN harvests the energy from the PS, it generally uses that energy to transmit the information. Therefore, an SN can harvest the energy from the PS but cannot forward that energy to other SNs. Accordingly, the communication technique is deployed in two-phase, energy transfer (ET) and information transfer (IT) (Fig. 2), as follows:

- In the first phase, during the time interval αT , S_m harvests energy from P_n . The harvested energy at S_m can be expressed as follows [33], [34]:

$$E_{P_n S_m} = \eta \alpha P_0 T \frac{|h_{P_n S_m}|^2}{d_{P_n S_m}^\theta}, \quad (1)$$

where α is a fraction of the total time block for which $0 < \alpha < 1$, and η is used to represent the energy

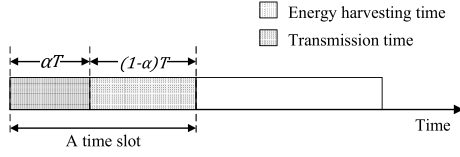


FIGURE 2. A considered total time block T is used for both energy harvesting and communication, in which the time αT is used to harvest energy from the multiple PSs, while the remaining time $(1 - \alpha)T$ is used to transmit the information to the BS.

conversion efficiency of an SN (which depends on the rectification process and the energy harvesting circuitry) where $0 \leq \eta \leq 1$. P_0 and T are the transmit power of the PSs and the total time block, respectively. The path loss exponent and the distance from P_n to S_m are expressed by θ and $d_{P_n S_m}$, respectively, and $|\cdot|$ is the absolute value operator.

- In the second phase, because different propagation effects can be suffered by the considered system communication, the SNs, jammer, BS, and ENs are generally subject to random fluctuations that impact the link capacities. In the considered model, the BS, jammer, and ENs can estimate the CSI of the SNs, while the SNs can determine the CSI of the BS, jammer, and the PSs based on the reciprocity of the wireless channel [35]. In addition, when the ENs are passive eavesdroppers (in which case their CSI cannot be estimated by the SNs, we can still consider that certain types of statistical CSI knowledge about the ENs can be estimated at the SNs such that we can set a worst-case value (best possible passive EN) to design the transmission [36] or detect inadvertently leaked local oscillator power from the receiver at the RF front-ends of the ENs [37].

C. POWER CONSTRAINT OF SNs

After the energy harvesting process, an SN also performs the CSI estimation to determine which PS is the best PS for its energy harvesting purposes. Here, the best PS is selected among N PSs, which allows S_m to use the maximal transmit power to improve the secrecy performance. Thus, the transmit power of S_m in the remaining time slot $(1 - \alpha)T$ and at the specific channel n is constrained by the harvested energy [38]; i.e.,

$$P_{P_n S_m} (1 - \alpha) T \leq E_{P^* S_m}, \quad (2)$$

where P^* is selected among the N PSs such that the channel power gain of the $P^* S_m$ link is largest; i.e.,

$$h_{P^* S_m} \triangleq \max_{n=1, \dots, N} \{|h_{P_n S_m}|\}. \quad (3)$$

By substituting (1) into (2), we can derive (4) as follows:

$$\begin{aligned} P_{P_n S_m} &\leq P_{P^* S_m} = \frac{E_{P^* S_m}}{(1 - \alpha) T} \\ &= \frac{\eta \alpha P_0}{(1 - \alpha) d_{P^* S_m}^\theta} |h_{P^* S_m}|^2, \end{aligned} \quad (4)$$

We can obtain the probability density function (PDF) and the cumulative distribution function (CDF) of the $\gamma_{P^* S_m}$, respectively, as follows [39]:

$$f_{\gamma_{P^* S_m}}(x) = \frac{N}{\lambda_{P^* S_m}} e^{-\frac{x}{\lambda_{P^* S_m}}} \left(1 - e^{-\frac{x}{\lambda_{P^* S_m}}}\right)^{N-1}, \quad (5)$$

$$F_{\gamma_{P^* S_m}}(x) = \left(1 - e^{-\frac{x}{\lambda_{P^* S_m}}}\right)^N, \quad (6)$$

where $\gamma_{P^* S_m} = \frac{|h_{P^* S_m}|^2}{d_{P^* S_m}^\theta}$, $\lambda_{P^* S_m} = \frac{\mathbb{E}(|h_{P^* S_m}|^2)}{d_{P^* S_m}^\theta}$, and $\mathbb{E}(\cdot)$ is the expectation operator.

D. INFORMATION TRANSMISSION

During the remaining duration of $(1 - \alpha)T$, the best SN S^* is selected among the M SNs such that the channel power gain of the S^* -to- B link is largest, i.e.,

$$h_{S^* B} \triangleq \max_{m=1, \dots, M} \{|h_{S_m B}|\}. \quad (7)$$

Accordingly, we can obtain the CDF of $\gamma_{S^* B}$ as follows:

$$F_{\gamma_{S^* B}} = \left(1 - e^{-\frac{x}{\lambda_{S^* B}}}\right)^M, \quad (8)$$

where $\lambda_{S^* B} = \frac{\mathbb{E}(|h_{S^* B}|^2)}{d_{S^* B}^\theta}$, and $d_{S^* B}$ is the distance from S^* to B .

S^* then transmits a signal $x(t)$ to the BS. During this time, the jammer also generates jamming signals $z(t)$ to the ENs and the BS [40], [41]. Thus, the signal received at the BS is given by

$$y(t) = \sqrt{\frac{P_{S^*}}{d_{S^* B}^\theta}} h_{S^* B} x(t) + \sqrt{\frac{P_J}{d_{JB}^\theta}} h_{JB} z(t) + n_B, \quad (9)$$

where d_{JB} is the distance from J to B , P_J represents the transmit power of J , and n_B is an additive white complex Gaussian noise at B , for which $n_B \in \mathcal{CN}(0, N_B)$, and we suppose that $N_B = N_0$.

We assume that the BS and the jammer cooperate with each other such that the jamming signal can be discarded at the BS. Here, both the BS and the jammer generate and index the same set of Gaussian pseudo-random jamming signals for synchronization. After the jammer transmits the actual signal to the BS, unlike the unknown ENs, the BS can disregard this signal using its prior information [42]; thus, the instantaneous received SNR at the BS is given by

$$\gamma_B = P_{S^*} \frac{|h_{S^* B}|^2}{N_0 d_{S^* B}^\theta} = \zeta \gamma_{P^* S^*} \gamma_{S^* B}, \quad (10)$$

where $\zeta = \frac{\eta \alpha P_0}{N_0 (1 - \alpha)}$, $\gamma_{P^* S^*} = \frac{|h_{P^* S^*}|^2}{d_{P^* S^*}^\theta}$, and $\gamma_{S^* B} = \frac{|h_{S^* B}|^2}{d_{S^* B}^\theta}$.

The multiple ENs can overhear the packet transmitted and can attempt to extract $x(t)$ without using an active attack. In addition, the ENs also receive interference signals from the jammer; hence, the received signal at E_k is denoted as follows:

$$z_k(t) = \sqrt{\frac{P_{S^*}}{d_{S^* E_k}^\theta}} h_{S^* E_k} x(t) + \sqrt{\frac{P_J}{d_{JE_k}^\theta}} h_{JE_k} z(t) + n_{E_k}, \quad (11)$$

where $d_{S^*E_k}$ and d_{JE_k} are the distance from S^* to E_k and from J to E_k , respectively, and n_{E_k} is additive white complex Gaussian noise. Assume that all n_{E_k} are equivalent and that $n_{E_k} \in \mathcal{CN}(0, N_{E_k})$. We also suppose that $N_{E_k} = N_0$.

Thus, the instantaneous received signal-to-interference-plus-noise ratio (SINR) at E_k is given by

$$\gamma_{E_k} = \frac{P_{P^*S^*} |h_{S^*E_k}|^2}{d_{S^*E_k}^\theta \left[\frac{P_J}{d_{JE_k}^\theta} |h_{JE_k}|^2 + N_0 \right]} = \zeta \gamma_{P^*S^*} \tilde{\gamma}_{E_k}, \quad (12)$$

where $\tilde{\gamma}_{E_k} = \frac{\gamma_{S^*E_k}}{\xi \gamma_{JE_k} + 1}$, $\gamma_{S^*E_k} = \frac{|h_{S^*E_k}|^2}{d_{S^*E_k}^\theta}$, $\gamma_{JE_k} = \frac{|h_{JE_k}|^2}{d_{JE_k}^\theta}$, and $\xi = \frac{P_J}{N_0}$.

From this, the CDF of $\tilde{\gamma}_{E_k}$ can be expressed as follows:

$$F_{\tilde{\gamma}_{E_k}} = \Pr \{ \tilde{\gamma}_{E_k} < x \} = \Pr \left\{ \frac{\gamma_{S^*E_k}}{\xi \gamma_{JE_k} + 1} < x \right\} = \int_0^\infty F_{\gamma_{S^*E_k}} [x (\xi \gamma_{JE_k} + 1)] f_{\gamma_{JE_k}} (\gamma_{JE_k}) d\gamma_{JE_k}, \quad (13)$$

where $\Pr\{\cdot\}$ is the probability.

After the mathematical manipulations, the expression of (13) can be approximated as follows:

$$F_{\tilde{\gamma}_{E_k}} = 1 - \frac{\lambda_{S^*E_k}}{\xi \lambda_{JE_k} + \lambda_{S^*E_k}} e^{-\frac{x}{\lambda_{S^*E_k}}} \approx 1 - e^{-\frac{\xi \lambda_{JE_k}}{\lambda_{S^*E_k}} x} e^{-\frac{x}{\lambda_{S^*E_k}}} = 1 - e^{-\frac{x}{\tilde{\lambda}_{E_k}}}, \quad (14)$$

where $\lambda_{S^*E_k} = \frac{\mathbb{E}(|h_{S^*E_k}|^2)}{d_{S^*E_k}^\theta}$, $\lambda_{JE_k} = \frac{\mathbb{E}(|h_{JE_k}|^2)}{d_{JE_k}^\theta}$, and $\tilde{\lambda}_{E_k} = \frac{\lambda_{S^*E_k}}{\xi \lambda_{JE_k} + 1}$. The approximation in (14) can be understood as ensuring that the interference from the jammer to the ENs is much larger than the background noise power, i.e., $\xi \gg 1$.

IV. SECRECY PERFORMANCE EVALUATIONS

A. CHANNEL CAPACITY

The instantaneous channel capacity of the S -to- B link without the interference signals from J can be defined as follows:

$$C_B = W \log_2(1 + \gamma_B), \quad (15)$$

where W is the system bandwidth.

The instantaneous channel capacity of the S -to- E_k link affected by the interfering signals from J can be defined as follows:

$$C_{E_k} = W \log_2(1 + \gamma_{E_k}). \quad (16)$$

Here, we assume a normalized bandwidth $W = 1$ as in the previous capacity definition. Because channel capacity is

a non-negative metric as defined in [43], we can obtain the instantaneous secrecy capacity at E_k as

$$C_{sec_k} = [C_B - C_{E_k}]^+ = \begin{cases} \log_2 \left(\frac{1 + \gamma_B}{1 + \gamma_{E_k}} \right), & \gamma_B > \gamma_{E_k} \\ 0, & \gamma_B \leq \gamma_{E_k}. \end{cases} \quad (17)$$

Under the conditions where the SNs are being served by P^* and the full location information is available for E_k , the joint CDF of γ_B and γ_{E_k} is given by

$$F_{\gamma_B, \gamma_{E_k}}(x, y) = \Pr \{ \gamma_B < x, \gamma_{E_k} < y \}. \quad (18)$$

By substituting Eqs. (10) and (12) into Eq. (18), we can rewrite this equation as follows:

$$F_{\gamma_B, \gamma_{E_k}}(x, y) = \Pr \{ \zeta \gamma_{P^*S^*} \gamma_{S^*B} < x, \zeta \gamma_{P^*S^*} \tilde{\gamma}_{E_k} < y \} = \Pr \left\{ \gamma_{S^*B} < \frac{x}{\zeta \gamma_{P^*S^*}}, \tilde{\gamma}_{E_k} < \frac{y}{\zeta \gamma_{P^*S^*}} \right\} = \int_0^\infty F_{\gamma_{S^*B}} \left(\frac{x}{\zeta z} \right) F_{\tilde{\gamma}_{E_k}} \left(\frac{y}{\zeta z} \right) f_{\gamma_{P^*S^*}}(z) dz. \quad (19)$$

Substituting Eqs. (5), (8), and (14) into Eq. (19), the joint CDF of γ_B and γ_{E_k} can be expressed as

$$F_{\gamma_B, \gamma_{E_k}}(x, y) = \int_0^\infty \left(1 - e^{-\frac{x}{\zeta \lambda_{S^*B} z}} \right)^M \left(1 - e^{-\frac{y}{\zeta \tilde{\lambda}_{E_k} z}} \right) \times \frac{N e^{-\frac{z}{\lambda_{P^*S^*}}}}{\lambda_{P^*S^*}} \left(1 - e^{-\frac{z}{\lambda_{P^*S^*}}} \right)^{N-1} dz. \quad (20)$$

Using Eqs. (A.1) and (A.2) in the appendix to solve the integral in Eq. (20), the function $F_{\gamma_B, \gamma_{E_k}}(x, y)$ can be calculated as follows:

$$F_{\gamma_B, \gamma_{E_k}}(x, y) = \widetilde{\sum}(M) \widetilde{\sum}(N) [\nu K_1(\nu) - \vartheta K_1(\vartheta)] + \widetilde{\sum}(N) [1 - \rho K_1(\rho)], \quad (21)$$

where $K_\sigma(\cdot)$ is the Bessel function with an imaginary argument, $\widetilde{\sum}(M) = \sum_{m=1}^M \frac{(-1)^{m+1} (M)!}{m! (M-m)!}$, $\widetilde{\sum}(N) =$

$$\sum_{n=0}^{N-1} \frac{(-1)^n (N)!}{(n+1)! (N-n-1)!}, \quad \nu = 2\sqrt{\frac{mx(n+1)}{\zeta \lambda_{P^*S^*} \lambda_{S^*B}}},$$

$$\vartheta = \sqrt{\frac{4(mx \tilde{\lambda}_{E_k} + \lambda_{S^*B} y)(n+1)}{\zeta \lambda_{P^*S^*} \lambda_{S^*B} \tilde{\lambda}_{E_k}}}, \text{ and } \rho = 2\sqrt{\frac{y(n+1)}{\zeta \lambda_{P^*S^*} \tilde{\lambda}_{E_k}}}.$$

B. EXISTENCE PROBABILITY OF SECRECY CAPACITY

In this subsection, we use a probabilistic metric to characterize the secure communication, which is referred to as PCS. This metric is also an important performance measurement and is useful for system designers to evaluate the secrecy performance of a wireless sensor communication system [44]. For a single E_k with PCS, the SNs can transmit a message safely over the fading channel if the instantaneous SNR of

the main channel is greater than that of the wiretap channel; in other words, when $C_{sec} > 0$, the system can take advantage of physical layer secrecy.

1) BEST-NODE SCHEDULING SCHEME

Assuming that all channels are independent, the probability of achieving a successful secure communication between S^* and B for a single E_k is defined as

$$PCS_k^{best} = \Pr \{C_{sec_k} > 0\} = \int_0^\infty \left[\frac{\partial F_{\gamma_B, \gamma_{E_k}}(x, y)}{\partial x} \right]_{y=x} dx. \quad (22)$$

By substituting Eq. (21) into Eq. (22), we obtain the PCS of the best-node scheduling scheme with E_k as follows:

$$PCS_k^{best} = \int_0^\infty \left[\frac{\frac{\partial \{ \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) [\nu K_1(\nu) - \vartheta K_1(\vartheta)] \}}{\partial x}}{\frac{\partial \{ \widetilde{\Sigma}(N) [1 - \rho K_1(\rho)] \}}{\partial x}} \right]_{y=x} dx \quad (23)$$

Using Eqs. (A.3) and (A.4) in the appendix, this integral can then be solved; thus, the function PCS_k^{best} in Eq. (23) yields

$$PCS_k^{best} = \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) \frac{2m(n+1)}{\zeta \lambda_{S^*B} \lambda_{S^*B}} \times \int_0^\infty [-K_0(\nu) + K_0(\vartheta)] dx = \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) \left[\frac{\lambda_{S^*B}}{\lambda_{S^*B} + m \tilde{\lambda}_{E_k}} \right]. \quad (24)$$

In the considered EH-WSNs with multiple ENs, the information can be transmitted confidentially only when the instantaneous SNR of the BS is greater than that of any EN (this implies that the instantaneous SNR of the BS is greater than the largest EN of K ENs); i.e.,

$$\gamma_E \triangleq \max_{k=1, \dots, K} \{\gamma_{E_k}\}. \quad (25)$$

Finally, based on Eqs. (24) and (25), in the considered EH-WSNs with multiple ENs, we can obtain the PCS for the best-node scheduling scheme as follows:

$$PCS^{best} = \Pr \left\{ \min_{1 \leq k \leq K} \{C_{sec_k}\} > 0 \right\} = \Pr \{C_{sec_1} > 0, C_{sec_2} > 0 \dots C_{sec_N} > 0\} = \prod_{k=1}^K \Pr \{C_{sec_k} > 0\} = \prod_{k=1}^K \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) \left[\frac{\lambda_{S^*B}}{\lambda_{S^*B} + m \tilde{\lambda}_{E_k}} \right]. \quad (26)$$

2) ROUND-ROBIN SCHEDULING SCHEME

For comparison purposes, we first study a conventional round-robin scheduling baseline scheme [29], where M SNs take turns accessing a given channel; thus, each SN has an equal opportunity to transmit its sensed data to the BS. Therefore, from Eq. (26), the PCS of the round-robin scheduling scheme of S_m with multiple ENs is obtained as follows:

$$PCS^{round_m} = \prod_{k=1}^K \widetilde{\Sigma}(N) \left[\frac{\lambda_{S_mB}}{\lambda_{S_mB} + \tilde{\lambda}_{E_k}} \right]. \quad (27)$$

Finally, in this scheme, the M SNs take turns transmitting their information to the BS, so the PCS of the considered EH-WSNs is the mean of the M SNs' PCS [15], yielding

$$PCS^{round} = \frac{1}{M} \sum_{m=1}^M PCS^{round_m} = \frac{1}{M} \sum_{m=1}^M \prod_{k=1}^K \widetilde{\Sigma}(N) \left[\frac{\lambda_{S_mB}}{\lambda_{S_mB} + \tilde{\lambda}_{E_k}} \right]. \quad (28)$$

C. SECRECY OUTAGE PROBABILITY

This subsection investigates the secrecy capacity of a Rayleigh fading channel in terms of the SOP. This metric is also an important performance measurement and is generally used to characterize the secrecy performance of a wireless communication system. The SOP is defined as the probability that the instantaneous secrecy capacity C_{sec} will fall below a predetermined secrecy rate threshold R (i.e., if $C_{sec} < R$, information security will not be guaranteed, and the system is said to be in outage; otherwise, perfect secrecy will be ensured).

1) BEST-NODE SCHEDULING SCHEME

The secrecy outage probability of the considered system for the best-node scheduling scheme with E_k is defined as follows [45]:

$$SOP_k^{best} = \Pr \{C_{sec_k}^{best} < R\} = \int_0^\infty \left[\frac{\partial F_{\gamma_B, \gamma_{E_k}}^{best}(x, y)}{\partial y} \right]_{x=\iota} dy, \quad (29)$$

where $\iota = 2^R(1+y) - 1$.

Similar to a previous subsection, by substituting Eq. (21) into Eq. (29), we obtain

$$SOP_k^{best} = \int_0^\infty \left[\frac{\frac{\partial \{ \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) [\nu K_1(\nu) - \vartheta K_1(\vartheta)] \}}{\partial y}}{\frac{\partial \{ \widetilde{\Sigma}(N) [1 - \rho K_1(\rho)] \}}{\partial y}} \right]_{x=\iota} dy. \quad (30)$$

We can solve this integral using Eqs. (A.1) to (A.4); hence, SOP_k^{best} can be calculated as follows:

$$SOP_k^{best} = 1 - \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) \left[\frac{\lambda_{S^*B}}{m \tilde{\lambda}_{E_k} 2^R + \lambda_{S^*B}} \beta K_1(\beta) \right], \quad (31)$$

where $\beta = 2\sqrt{\frac{m(n+1)(2^R-1)}{\zeta\lambda_{p^*S^*}\lambda_{S^*B}}}$.

Finally, in the considered EH-WSNs with multiple ENs, the SOP for the best-node scheduling scheme becomes

$$\begin{aligned}
 SOP^{best} &= \Pr \left\{ \max_{1 \leq k \leq K} \{C_{sec_k}^{best}\} < R \right\} \\
 &= 1 - \Pr \left\{ \min_{1 \leq k \leq K} \{C_{sec_k}^{best}\} \geq R \right\} \\
 &= 1 - \prod_{k=1}^K \Pr \{C_{sec_k}^{best} \geq R\} \\
 &= 1 - \prod_{k=1}^K \left[1 - \Pr \{C_{sec_k}^{best} < R\} \right] \\
 &= 1 - \prod_{k=1}^K \left\{ \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) \left[\frac{\lambda_{S^*B}}{m\tilde{\lambda}_{E_k}2^R + \lambda_{S^*B}} \beta K_1(\beta) \right] \right\}. \quad (32)
 \end{aligned}$$

2) ROUND-ROBIN SCHEDULING SCHEME

Similar to a PCS of the round-robin scheduling scheme, the SOP of the considered system for the round-robin scheduling scheme at S_m with multiple ENs can be expressed as follows:

$$SOP^{round_m} = 1 - \prod_{k=1}^K \left[\widetilde{\Sigma}(N) \frac{\lambda_{S_mB}}{\tilde{\lambda}_{E_k}2^R + \lambda_{S_mB}} \beta' K_1(\beta') \right], \quad (33)$$

where $\beta' = 2\sqrt{\frac{(n+1)(2^R-1)}{\zeta\lambda_{p^*S_m}\lambda_{S_mB}}}$.

Finally, in this scheme, the SOP of the considered system becomes

$$\begin{aligned}
 SOP^{round} &= \frac{1}{M} \sum_{m=1}^M SOP^{round_m} \\
 &= \frac{1}{M} \sum_{m=1}^M \left\{ 1 - \prod_{k=1}^K \left[\widetilde{\Sigma}(N) \frac{\lambda_{S_mB}}{\tilde{\lambda}_{E_k}2^R + \lambda_{S_mB}} \beta' K_1(\beta') \right] \right\}. \quad (34)
 \end{aligned}$$

D. ASYMPTOTIC PERFORMANCE ANALYSIS WITH HIGH-SNR-REGIME

The closed-form expression for the SOP in Eq. (22) relies on the Bessel function, i.e., $K_1(\cdot)$. Therefore, we can obtain the approximate expression $xK_1(x)$ when $x \rightarrow 0$ as follows [46]:

$$xK_1(x) \stackrel{x \rightarrow 0}{\approx} 1 + \frac{x^2}{2} \left[\ln\left(\frac{x}{2}\right) + \varsigma - \frac{1}{2} \right] \triangleq K_{asympt}(x), \quad (35)$$

where ς is the Euler-Mascheroni constant, and $K_{asympt}(x)$ are the approximate expressions for $xK_1(x)$ as $x \rightarrow 0$. Due to the relation $\zeta = \frac{\eta\alpha P_0}{N_0(1-\alpha)}$, we have

$$\lim_{\frac{N_0}{P_0} \rightarrow 0} SOP = \lim_{\zeta^{-1} \rightarrow 0} SOP. \quad (36)$$

Applying Eqs. (35) and (36) to Eqs. (32) and (34), the asymptotic expression of the SOP for best-node scheduling and round-robin scheduling schemes at very high ζ (i.e., $\zeta^{-1} \rightarrow 0$) are expressed, respectively, as follows:

$$\begin{aligned}
 SOP_{asympt}^{best} &= \lim_{\zeta^{-1} \rightarrow 0} SOP^{best} \\
 &= 1 - \prod_{k=1}^K \left\{ \widetilde{\Sigma}(M) \widetilde{\Sigma}(N) \left[\frac{\lambda_{S^*B}}{m\tilde{\lambda}_{E_k}2^R + \lambda_{S^*B}} K_{asympt}(\beta) \right] \right\}, \quad (37)
 \end{aligned}$$

and

$$\begin{aligned}
 SOP_{asympt}^{round} &= \lim_{\zeta^{-1} \rightarrow 0} SOP^{round} \\
 &= \frac{1}{M} \sum_{m=1}^M \left\{ 1 - \prod_{k=1}^K \left[\widetilde{\Sigma}(N) \frac{\lambda_{S_mB}}{\tilde{\lambda}_{E_k}2^R + \lambda_{S_mB}} K_{asympt}(\beta') \right] \right\}. \quad (38)
 \end{aligned}$$

V. NUMERICAL RESULTS AND DISCUSSION

In this section, we provide numerical examples of the secrecy performance of EH-WSNs with a friendly jammer and multiple ENs based on two system metrics: PCS and SOP. Specifically, we investigate these two metrics by considering the effects of energy harvesting time α , number of PSs N , number of SNs M , number of ENs K , distance from the SN to the BS d_{SB} , power of the jammer P_J , and signal-to-noise ratio SNR. Unless otherwise specified, the following system parameters are used for both the analysis and the simulation [45]–[48]:

- The secrecy outage threshold $R = 1$ Kbps.
- The path loss exponent $\theta = 3$.
- $N = \{1, 3\}$, $M = \{1, 10\}$, and $K = \{1, 3\}$.

Figs. 3 and 4 depict the PCS of the system versus the distance from the SN to the BS $d_{SB} \in (0, 2)$ for the M SNs and the K ENs, respectively. Fig. 3 shows a comparison between the closed-form PCS of the round-robin scheduling scheme and the best-node scheduling scheme for different numbers of SNs M . We can see that the PCS^{round} corresponding to $M = 2$ is the same as that corresponding to $M = 5$ and $M = 10$, further showing that no security benefit is achieved by the round-robin scheduling as the number of SNs increases.

We can also observe that the PCS^{best} can be improved by increasing M or decreasing K . This is because the use of a higher M outperforms the use of a lower M because the diversity gain increases at the SNs. Note that when there are more eavesdroppers that can receive more information, their abilities to decode the secret message are improved. Further, we can also readily observe the jammer effects on the PCS (it increases when we use a jammer) which confirms the improvement in the secrecy performance of EH-WSNs from a jammer. Moreover, we can observe from these figures that

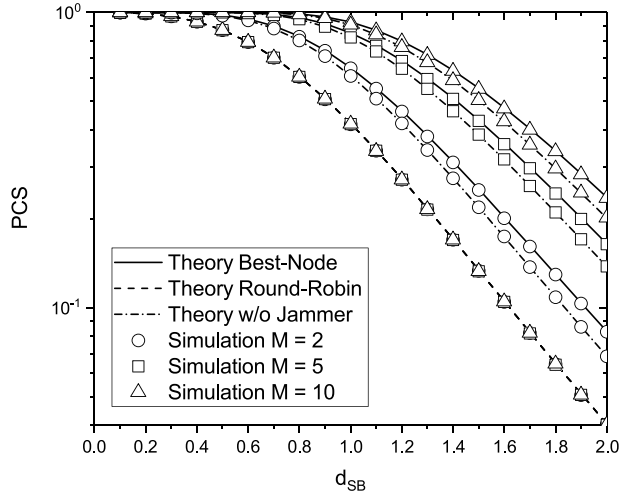


FIGURE 3. Impact of various d_{SB} and various numbers of SNs M on the existence probability of secrecy capacity with $N = 2$, $K = 2$, $\alpha = 0.5$, $\eta = 0.85$, $P_J = 15$ dB, and $P_0/N_0 = 18$ dB.

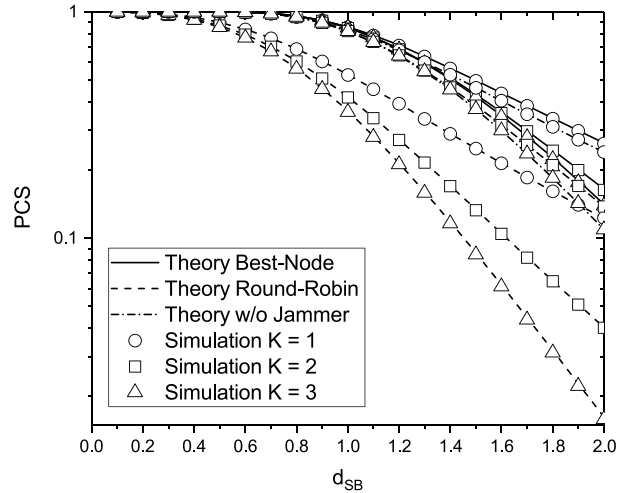


FIGURE 4. Impact of various d_{SB} and various numbers of ENs K on the existence probability of secrecy capacity with $N = 2$, $M = 5$, $\alpha = 0.5$, $\eta = 0.85$, $P_J = 15$ dB, and $P_0/N_0 = 18$ dB.

the PCS of the best-node scheduling scheme is higher than that of the round-robin scheduling scheme.

Figs. 5 and 6 demonstrate the effect of the fraction of the total time block $\alpha \in (0, 1)$ and the energy conversion efficiency of the SN $\eta \in (0, 1)$ on the SOP when $K = 2$, $P_J = 15$ dB, and $P_0/N_0 = 18$ dB. It can be observed that the SOP of the system can be improved by increasing either α or η . This is because higher α and η values means that more energy is harvested at the SN for the ET phase and that less time will be allocated for the IT phase. Based on Eq. (4), energy harvesting at the SN will increase as α or η increases; i.e., the more time the system has for harvesting energy, the more energy can be harvested at the SN to make the system more secure, effectively reducing the SOP. Based on the numerical results, we also observe that the SOP_{round} is the same under $M = 2$, $M = 5$, and $M = 10$ in both

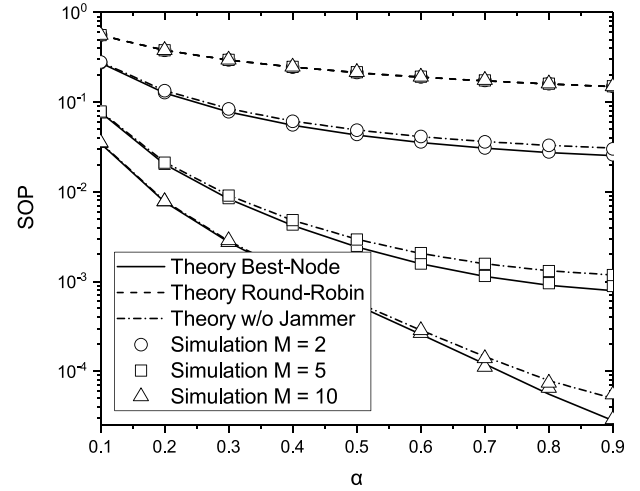


FIGURE 5. Impact of various fractions of the total time block α and various numbers of SNs M on the secrecy outage probability with $K = 2$, $N = 2$, $\eta = 0.85$, $P_J = 15$ dB, and $P_0/N_0 = 18$ dB.

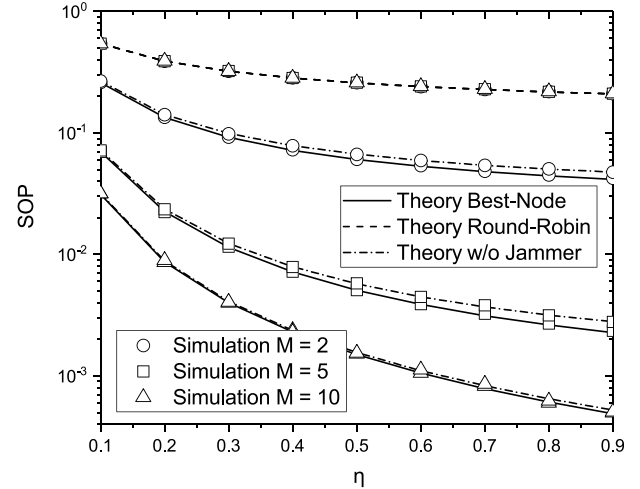


FIGURE 6. Impact of various energy conversion efficiencies of the SN η and various numbers of SNs M on the secrecy outage probability with $K = 2$, $N = 2$, $\alpha = 0.5$, $P_J = 15$ dB, and $P_0/N_0 = 18$ dB.

Figs. 5 and 6, which confirms that the secrecy performance is not affected by the number of SNs in a round-robin scheduling scheme.

Fig. 7 plots the SOP of the considered EH-WSNs versus $P_J = \{0, 20\}$ dB for various N . We observe that the SOP of the system with the jammer increases slightly with increasing P_J but that the SOP of the system without the jammer is the same for all P_J . This is because the jammer generates signals that interfere with the ENs; hence, the higher power of the jammer improves the security performance of the system. Note that the secrecy performance becomes more pronounced as the number of PSs increases.

Figs. 8 illustrates the effect of the SNR at the SNs on the SOP. We can see that the secrecy performance of the best-node scheduling scheme is better than that of the other schemes for all SNRs and numbers of

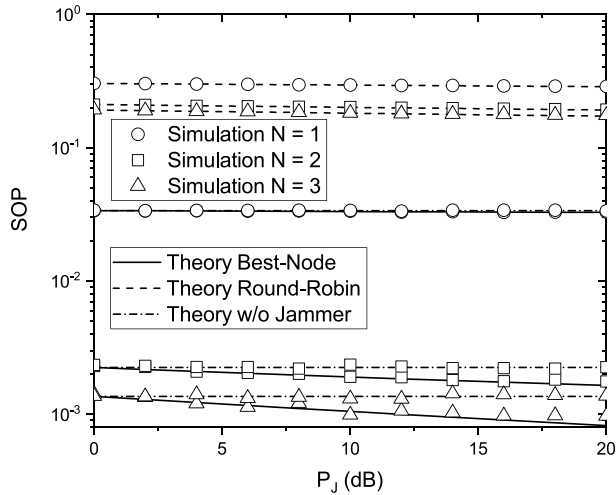


FIGURE 7. Impact of various P_J and various numbers of PSs N on the secrecy outage probability with $K = 2$, $M = 5$, $\alpha = 0.5$, $\eta = 0.85$, and $P_0/N_0 = 18$ dB.

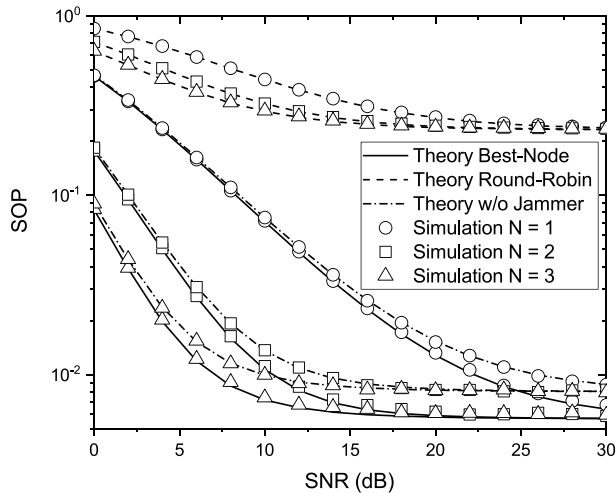


FIGURE 8. Impact of various SNRs P_0/N_0 and various numbers of PSs N on the secrecy outage probability with $K = 2$, $M = 5$, $\alpha = 0.5$, $\eta = 0.85$, and $P_J = 15$ dB.

PSs, which confirms the usefulness of our closed-form solution.

Figs. 9 and 10 depict the SOP versus the SNR of the main channel with various M and N . These figures verify the agreement between the analysis and simulation, while the asymptotic analysis is very close to the analysis in the high-SNR regime. The asymptotic result for the system obviously approximates the corresponding simulation result in the high-SNR regime for various values of M and N . This validates our asymptotic results derived for the SOP: the SOP increases with increasing M and N . This implies that increasing the number of Ps and SNs significantly improves the secrecy outage probability, especially in the medium- and high-SNR regimes. Thus, from these figures, we can conclude that the best-node scheduling scheme significantly improves both the PCS and the SOP.

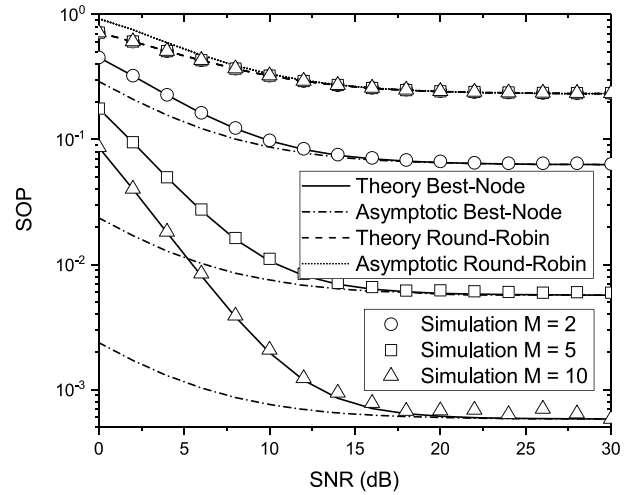


FIGURE 9. Impact of various SNRs P_0/N_0 and various numbers of SNs M on the secrecy outage probability and asymptotic results with $K = 2$, $N = 2$, $\alpha = 0.5$, $\eta = 0.85$, and $P_J = 15$ dB.

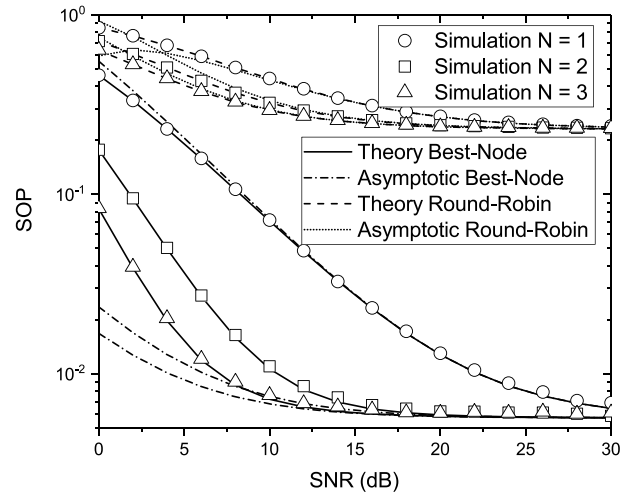


FIGURE 10. Impact of various SNRs P_0/N_0 and various numbers of PSs N on the secrecy outage probability and asymptotic results with $K = 2$, $M = 5$, $\alpha = 0.5$, $\eta = 0.85$, and $P_J = 15$ dB.

VI. CONCLUSION

In this paper, we derived the closed-form expression of the existence probability of the secrecy capacity and the secrecy outage probability of EH-WSNs with a jammer, multiple power stations, multiple sources, and a base station in the presence of multiple eavesdroppers. In addition, the best-node scheduling scheme was proposed to enhance system secrecy performance against eavesdropping attacks. We investigated two metrics, the PCS and the SOP, by considering the effects of various indicators of the system to clarify our secrecy performance analysis. In addition, we further demonstrated the accuracy of the analysis using Monte-Carlo simulations. For future work, we are in the process of evaluating a system with multiple source nodes and multiple jammers with relaying techniques to illustrate a practical implementation of EH-WSNs.

APPENDIX

$$\int_0^{\infty} e^{-px} dx = \frac{1}{p} \quad s.t. [Rep > 0]. \quad (A.1)$$

$$\int_0^{\infty} x^{v-1} e^{-\frac{\beta}{x} - \gamma x} dx = 2 \left(\frac{\beta}{\gamma} \right)^{\frac{v}{2}} K_v \left(2\sqrt{\beta\gamma} \right) \quad s.t. [Re\beta > 0, Re\gamma > 0]. \quad (A.2)$$

$$\left(\frac{d}{xdx} \right)^m \{x^v K_v(x)\} = (-2)^m x^{v-m} K_{v-m}(x). \quad (A.3)$$

$$\int_0^{\infty} x^{\mu} K_v(ax) dx = 2^{\mu-1} a^{-\mu-1} \Gamma \left(\frac{1+\mu+v}{2} \right) \times \Gamma \left(\frac{1+\mu-v}{2} \right) \quad s.t. [Re(\mu+1 \pm v) > 0, Rea > 0]. \quad (A.4)$$

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